Geometric series

1 $20\,475$

2

a 14.29

b -204.92

3 683

4 $\dfrac{5461}{64}$

5 17

6 9.143

7

a $\dfrac{1}{2}$

b 10.0

8

a $x\left(\dfrac{1-\left(\dfrac{2}{5}x\right)^n}{1-\dfrac{2}{5}x}\right)$

b 2555

9 $\dfrac{1-4\times\left(\dfrac{1}{4}\right)^n+3\times\left(\dfrac{1}{5}\right)^n}{12}$

10

a

i $n=10$ **ii** $S_{10}=-1364$

b

i $n=6$ **ii** $S_6=\dfrac{1365}{16}$

11

a $n=8$ **b** $n=9$ **c** $n=6$ **d** $n=6$

12 $r=3$

13 $a=327\,680$

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14 $7\,174\,449$

Infinite series

15 The absolute value of the common ratio must be less than 1.

16

a i $r = 2$ ii Divergent

b i $r = -4$ ii Divergent

c i $r = \frac{1}{2}$ ii Convergent

d i $r = \frac{1}{4}$ ii Convergent

17

a i $r = \frac{1}{4}$ ii $\frac{8}{3}$

b i $r = \frac{1}{5}$ ii $\frac{625}{4}$

c i $r = -\frac{1}{2}$ ii $\frac{32}{3}$

d i $r = -\frac{1}{4}$ ii $\frac{112}{5}$

18

a $r = \frac{1}{2}$

b $2, \frac{1}{2}, \frac{1}{4}$

19 $x < 8$ and $x > 6$

20 $x = \frac{1}{5}$

21 $576, 144, 36$

22

a $n = 11$

b 9

23

a $r = \frac{\sqrt{5}}{5}$

b $\frac{\left(25 + 5\sqrt{5}\right)}{4}$

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24 $\frac{1}{3}$

Sigma notation

25

a 340 **b** 7205 **c** 31 **d** 4914

e -252 **f** 242 **g** $\frac{3267}{4}$ **h** $-\frac{40}{27}$

26 $\sum_{k=1}^{11}\left(-\frac{1}{2}\right)^{k-1}$

27

a 4 **b** 4 **c** $\frac{1}{12}$ **d** $-\frac{7}{4}$

e 9 **f** $\frac{1}{8}$

Repeating decimals

28

a $\frac{8}{10}+\frac{8}{100}+\frac{8}{1000}+\frac{8}{10\,000}+\frac{8}{100\,000}$ **b** $a=\frac{8}{10}, r=\frac{1}{10}$

c $\frac{8}{9}$

29

a $\frac{44}{100}$ **b** $\frac{44}{10\,000}$

c $\frac{44}{1\,000\,000}$ **d** $a=\frac{44}{100}, r=\frac{1}{100}$

e $\frac{4}{9}$

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30

a

i $0.6 + 0.06 + 0.006 + 0.0006 + \ldots$

ii $\frac{2}{3}$

b

i $0.06 + 0.006 + 0.0006 + 0.00006 + \ldots$

ii $\frac{1}{15}$

31

a $0.252525\ldots$

b $\frac{25}{99}$

32 $\frac{2}{9}$

33 $\frac{19}{33}$

34 $\frac{994}{999}$

Applications

35

a 0.04

b $5000 \times 1.04 + 700$

c $5000 \times (1.04)^2 + 700 \times 1.04 + 700$

d $5000 \times (1.04)^3 + 700 \times (1.04)^2 + 700 \times 1.04 + 700$

e $5000 \times (1.04)^n + \frac{700\left((1.04)^n - 1\right)}{1.04 - 1}$

f $\$13\,293$

36

a

i $50\,000 \times 1.059 + 3000$

ii $50\,000 \times (1.059)^2 + 3000 \times 1.059 + 3000$

iii $50\,000 \times (1.059)^3 + 3000 \times (1.059)^2 + 3000 \times 1.059 + 3000$

b $\frac{3000\left((1.059)^n - 1\right)}{0.059}$

c $\$99\,791$

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37

a $A_1 = 30\,000 \times 1.015 - 600$

b

i $A_2 = 30\,000 \times (1.015)^2 - (1.015 \times 600) - 600$

ii $A_3 = 30\,000 \times (1.015)^3 - \left(1.015^2 \times 600\right) - (1.015 \times 600) - 600$

c $A_n = 30\,000 \times (1.015)^n - \dfrac{600\left((1.015)^n - 1\right)}{1.015 - 1}$

d $\$22\,908.60$

38

a $1400 \times (1.005)^n + \dfrac{165\left((1.005)^n - 1\right)}{(1.005) - 1}$ **b** $\$8165.81$

39

a $A_1 = 190\,000 \times 1.07 - x$

b

i $A_2 = 190\,000 \times (1.07)^2 - 1.07x - x$

ii $A_3 = 190\,000 \times (1.07)^3 - (1.07)^2\, x - 1.07x - x$

c $190\,000 \times (1.07)^n - \dfrac{x\left((1.07)^n - 1\right)}{0.07}$

d $\$17\,934.66$

40

a $110\,000 \times 1.04^n - \dfrac{x\left(1.04^n - 1\right)}{0.04}$ **b** $x = 4488.88$

41

a $190\,000 \times 1.002\,5^n - \dfrac{x\left(1.002\,5^n - 1\right)}{0.002\,5}$ **b** $x = 801.05$

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